MIRAI BANK PROJECT

Milestone 1 [A1]

Blockchain Functionality Research Report

Table of Contents

1. Research Overview 3

2. Research into NFT Standards (Corresponding to: A1-1) 4

2.1 Premise: The Basic Architecture of Cardano NFTs 4

2.2 The Cardano Blockchain and the Structure of NFT Records 6

2.3 The Structure of NFT Metadata (Examples) 7

2.4 Comparison of CNFT Standards (CIP-25 / CIP-68) 10

2.5 Selected Standard and Rationale (Discussion) 12

3. Research into the Persistence Model for Decentralized Storage (Corresponding to: A1-2) 14

3.1 Overview of Decentralized Storage 14

3.2 Basic Comparison of IPFS and Arweave 15

3.3 Supplementary Note: Compatibility Between NFT Standards and Storage 17

3.4 Selected Storage and Rationale (Discussion) 18

4. Summary 20

1. Research Overview

This report is Milestone 1 ("Prototype research and basic specifications") Output A1, the Blockchain Functionality Research Report. ChainCapsule is a service built around the concept of "accurately recording people's thoughts for the future," whose greatest value lies in protecting the records entrusted to it by users, in a form that no one can alter, until the specified future date. How to technically achieve this property of being "unable to be rewritten" is one of the most important design decisions determining the trustworthiness of this service.

This report therefore investigates and compares two technical elements that form the foundation of that trustworthiness: a mechanism to prove ownership and authenticity (NFTs), and a mechanism to keep recorded content stored for a long period of time (decentralized storage). Rather than simply listing technical pros and cons, we have tried to explain as carefully as possible why each technology suits a service like ChainCapsule, and conversely what constraints or trade-offs must be accepted. Specifically, the research covers the following two points.

- **A1-1:**

  Research into the use of NFTs to define ownership (comparison and selection of Cardano NFT standards)
- **A1-2**

  Comparative evaluation of persistence models for decentralized storage (comparison and selection between IPFS and Arweave)

Chapter 2 organizes the content corresponding to A1-1, Chapter 3 covers A1-2, and Chapter 4 provides an overall summary of both.

2. Research into NFT Standards (Corresponding to: A1-1)

2.1 Premise: The Basic Architecture of Cardano NFTs

An NFT (Non-Fungible Token) is a kind of digital certificate that can prove on the blockchain that something is one of a kind, in the entire world. In ChainCapsule, an NFT is issued the moment a capsule is finalized (sealed), giving it the role of recording "when, by whom, and what content was recorded" in a form that cannot later be altered.

Ethereum is widely known as a blockchain that handles NFTs, but Cardano, which this project adopts, realizes NFTs in a fundamentally different way from Ethereum. On Ethereum, NFTs are issued by writing small programs called "smart contracts," one at a time. This approach offers a high degree of flexibility, but the fee (gas cost) needed to run the program can spike depending on network congestion, and a bug in the program can directly translate into a defect in the asset itself.

On Cardano, by contrast, NFTs are issued as a type of asset called a "native asset," a standard capability that the Cardano blockchain itself provides. There is no need to write a new special-purpose program; an NFT can be issued simply by applying the ordinary mechanism used for sending payments. As a result, fees are small and stable, and the risk of a record being broken by a program defect is also low. For a service like ChainCapsule, whose purpose is to "stably protect a finalized record over a long period of time," this combination of low cost and low risk is a major advantage. Table 2-1 summarizes the architectural differences between the two.

Item	Ethereum NFT	Cardano NFT (CNFT)
Blockchain model	Account-based	EUTXO (Extended UTXO)
NFT implementation	Smart contract (ERC-721 / ERC-1155)	Native asset (CIP-25 / CIP-68)
Metadata	Contract + off-chain	On-chain
Transaction fees	Gas fees tend to be high	Consistently low cost

Table 2-1 Architecture comparison between Ethereum NFTs and Cardano NFTs

As this comparison shows, Cardano's native-asset mechanism aligns well with the direction ChainCapsule is aiming for: "stably protecting records over a long period of time." The next section looks at exactly "where" on the blockchain, and "in what form," that NFT is actually recorded.

2.2 The Cardano Blockchain and the Structure of NFT Records

Before examining NFT standards such as CIP-25 and CIP-68, it is worth understanding exactly "where" within the blockchain information about an NFT is recorded in the first place. This section organizes the basic structure of the Cardano blockchain.

The Cardano blockchain is made up of a large number of "blocks" linked together like a chain. Each block consists broadly of two parts: a "header" and a "body." The header acts like the block's ID card, recording information such as when the block was created and which block it follows. In particular, by holding the "hash of the previous block" (a value like a fingerprint), any attempt to later alter an earlier block is guaranteed to leave a trace. This is the reason blockchains are said to be "tamper-resistant." The body, meanwhile, collectively records the actual transactions that took place, such as payments and NFT issuance.

What matters more for ChainCapsule is the content of these "transactions." Cardano adopts a model called EUTXO (Extended UTXO), and a single transaction is mainly composed of the following elements.

- **Inputs: information indicating which prior transaction the assets about to be used came from**
- **Outputs: information indicating to whom (which address) the assets are being sent**
- **Amount: the quantity of ADA, or of a custom token such as an NFT, being sent**
- **Datum: supplementary state data associated with a smart contract**
- **Witness (signature): data proving that the transaction was carried out by its rightful owner**
- **Metadata: arbitrary descriptive information attached to the transaction. A CIP-25-format NFT records its content mainly in this field**

In other words, a CIP-25 NFT writes its information into the transaction's "metadata" field, while a CIP-68 NFT writes into the "datum" field — this is the key difference between the two. This difference directly determines whether or not the metadata can later be updated, as discussed below.

2.3 The Structure of NFT Metadata (Examples)

This section shows, with concrete CIP-25 and CIP-68 examples, what kind of information is actually recorded in the metadata when ChainCapsule issues an NFT. There is no need to memorize the exact notation; the goal is simply to get a feel for "what information" is recorded and "in what form."

The CIP-25 Case

Under CIP-25, the transaction metadata used to issue an NFT is written directly in a JSON format (a general-purpose way of recording data as name/value pairs), as shown below.

```
{
  "721": {
    "<policy_id>": {
      "MiraiNFT001": {
        "name": "Mirai NFT project name",
        "image": "ipfs://JpegIpfsHash",
        "mediaType": "image/jpeg",
        "description": "Mirai NFT project description",
        "attributes": {
          "createdate": "2026/01/01",
          "opendate": "2036/01/01"
        },
        "files": [
          {
            "mediaType": "application/pdf",
            "src": "ipfs://PdfIpfsHash"
          }
        ]
      }
    },
    "version": "1.0"
  }
}
```

Figure 2-1 Example of CIP-25 metadata

"721" is the label number marking this as CIP-25; below it comes an identifier for the issuing entity (policy_id), and below that, information for each NFT name (MiraiNFT001 in

this example). "name" is the NFT's name, "image" is the location of the stored image (on IPFS, in this project's case), and "attributes" can freely hold custom supplementary information such as the creation date and planned opening date. "files" allows multiple additional files besides the image (for example, a letter in PDF format) to be attached. As this shows, CIP-25 is characterized by intuitive, easy-to-understand fields and a low barrier to implementation.

The CIP-68 Case

Under CIP-68, metadata is not written into the transaction's metadata field; instead, it is recorded in the "datum" of a special NFT called the "Reference NFT," in a format such as the following.

```
{
  "constructor": 0,
  "fields": [
    {
      "map": [
        { "k": {"bytes": "6e616d65"}, "v": {"bytes": "4d697261694e4654..."} },
        { "k": {"bytes": "696d616765"}, "v": {"bytes": "697066733a2f2f..."} }
      ]
    },
    { "int": 1 }
  ]
}
```

Figure 2-2 Example of a CIP-68 datum (Reference NFT)

Both the field names, such as "name" and "image," and their values are converted into hexadecimal byte strings for the computer to handle. This is somewhat harder for a human to read at a glance, but this structure is precisely what allows a smart contract program to rewrite the datum safely, following a fixed procedure. To make such a rewrite, the process takes the transaction containing the existing Reference NFT as an input, attaches a new datum, and sends the output back to the same address.

About the Policy ID (the Issuer's Identifier)

Every NFT is assigned a "Policy ID," an identifier for the entity that issued it. Under CIP-25, this Policy ID is basically derived from the address of the wallet used to issue the NFT, and

can optionally be combined with a mechanism that restricts the "period during which minting is allowed" (a time lock). By limiting this period, it becomes possible to guarantee that "no more of this NFT will ever be minted."

Under CIP-68, the Policy ID is calculated from the content of a small program (the minting script) used to issue and update the NFT. Where no update is planned, a simple approach that mints using only a specific wallet's signature can be used; where dynamic updates are involved, a more advanced program (a Plutus script) is used instead. Because ChainCapsule is designed, as described below, not to perform updates, this point is also a factor in the standard-selection decision.

2.4 Comparison of CNFT Standards (CIP-25 / CIP-68)

With the above as background, this section compares Cardano's NFT standards. Cardano's NFT standards fall broadly into two families: "CIP-25 (legacy NFT metadata)" and "CIP-68 (updatable reference NFT)." Table 2-2 provides an overview comparison of the two standards.

Item	CIP-25 (Legacy NFT metadata)	CIP-68 (Updatable reference NFT)
Purpose	NFT metadata standard	Updatable metadata + reference NFT
Metadata location	Transaction metadata (label 721)	Reference NFT's datum (on-chain)
Metadata update	Not possible (requires re-mint / additional mint / burn + re-mint)	Possible (rewrite the reference NFT's datum)
Compatibility	Widely supported by wallets and marketplaces alike	Newer-generation standard; support is increasing
Use cases	Simple NFTs, RWAs that do not require changes	Dynamic NFTs, games, RWAs, SFTs

Table 2-2 Overview comparison of CIP-25 and CIP-68

Technical Advantages and Disadvantages of CIP-25

Advantages	Disadvantages
• The most widely supported NFT standard on Cardano, supported by nearly all wallets and marketplaces • Can be implemented simply by writing JSON metadata at mint time • Well suited to NFTs that do not need to change, such as artwork, certificates, and RWAs with fixed information	• Metadata cannot be updated (updating requires a re-mint, an additional mint, or a burn plus re-mint) • Because it does not use a datum, integration with smart contracts is weak

Technical Advantages and Disadvantages of CIP-68

Advantages	Disadvantages
• Metadata can be updated simply by rewriting the reference NFT's datum (the User Token stays with its holder) • Separates the User Token (the asset itself) from the Reference NFT (its metadata), allowing flexible support for large-scale issuance, game items, RWAs, and more • Because the datum can hold structured data, it integrates well with smart contract logic • Also supports SFTs (semi-fungible tokens)	• Implementation is more complex because it involves managing two tokens: the Reference NFT and the User Token • Wallet and marketplace support is not yet as complete as for CIP-25 • Because it uses a datum, UTXO management must be considered

2.5 Selected Standard and Rationale (Discussion)

Building on the comparison so far, this section considers which NFT standard ChainCapsule should adopt. The most important question in making this decision was: "given the nature of the ChainCapsule service, is there truly a need to update an NFT's content after it has been issued?"

ChainCapsule is a service that preserves a record from a point in time chosen by the user, unchanged, until a specified future date, and reveals it only at the moment of opening. In other words, the very fact that "content cannot be rewritten" is the value of the service itself, so CIP-68's key strength — "metadata can be updated later" — can be seen as unnecessary for this service, or even a feature that should deliberately be avoided.

There is also a large difference from the standpoint of ease of implementation. CIP-25 is complete once JSON-format metadata is written into the issuing transaction, whereas CIP-68 requires managing two kinds of tokens together — the Reference NFT and the User Token — which increases the complexity of implementation and operation. In prototype development, this added complexity translates directly into longer development time and greater risk of bugs, so it must be evaluated carefully. In addition, when it comes to the state of the surrounding ecosystem, such as wallets and marketplaces, CIP-25 is the most widely adopted standard on Cardano, which also gives users the reassurance of being able to check their NFT in an ordinary wallet.

Selected Standard	**Adopt CIP-25 (the legacy NFT metadata standard)**
Rationale	**The primary purpose of the NFTs issued by this project is to serve as a certificate at the time a capsule is finalized, and there is no need to update the metadata after issuance. CIP-25 can be implemented simply by writing JSON metadata into the mint transaction, and it has the broadest wallet and marketplace support. CIP-68, with its update capability, is oriented toward dynamic use cases such as updating the valuation of game items or RWAs, and would introduce excessive complexity relative to this project's requirement (never changing a finalized record). For these reasons, CIP-25 is selected, prioritizing ease of implementation and compatibility.**

Note that this selection is based on the premise that update functionality will never be needed in the future either. If a future service is considered in which content changes dynamically (for example, a type of capsule that continuously adds a record of growth over time), a move to CIP-68, or a use-case-dependent mix of standards, would need to be reconsidered at that time.

3. Research into the Persistence Model for Decentralized Storage (Corresponding to: A1-2)

3.1 Overview of Decentralized Storage

While the NFT itself is recorded on the blockchain, writing the full body of the images or text that the NFT points to directly onto the blockchain would incur enormous cost. For this reason, it is common practice to store the body of the content in a separate mechanism called "decentralized storage," and to have the NFT record only information indicating where that content is stored. Decentralized storage is a technology that, rather than concentrating data on a single server, distributes it across multiple servers or computers. This mechanism improves data reliability and availability, and provides the benefit that data can still be recovered from other servers even if a particular server fails.

Decentralized storage offers benefits such as fault tolerance (data loss can be prevented even if one server fails), scalability (capacity can easily be expanded by adding servers as needed), and fast access (data can be retrieved from a server near the user), but it also carries challenges such as management complexity, additional cost to maintain data consistency, and security risks associated with managing multiple nodes. For ChainCapsule, the choice of decentralized storage greatly affects how long, and how reliably, recorded content will continue to survive.

3.2 Basic Comparison of IPFS and Arweave

As destinations for storing the NFT body and related files, we compared two representative decentralized storage systems, IPFS and Arweave. Table 3-1 shows a comparison of their basic concepts.

Item	IPFS	Arweave
Architecture	P2P distributed file system	Blockchain (Blockweave)
Data identification	Content address (CID)	Transaction ID (permanent)
Nature of storage	Not permanent by default (pinning required)	Permanent storage (guaranteed by economic incentives)
Primary use	Fast delivery, flexible decentralized storage	Permanent archiving, NFT metadata, the Permaweb

Table 3-1 Basic concept comparison between IPFS and Arweave

Technical Advantages and Disadvantages of IPFS

Advantages	Disadvantages
• Content addressing (CID) makes tampering easy to detect • Content can be retrieved in parallel from nearby nodes, enabling fast delivery • The most standard destination for NFT images, with a rich ecosystem of development tools • Highly flexible, well suited to dynamic use cases and temporary data distribution	• Persistence is not guaranteed (data that is not pinned can be lost over time) • There is no built-in incentive design for node operators, leading to reliance on paid pinning services such as Pinata

Technical Advantages and Disadvantages of Arweave

Advantages	Disadvantages
• The one-time payment model guarantees, in theory, permanent data storage for over 200 years • Economic incentives through the Endowment provide continuity that does not depend on any single service provider • High tamper resistance through the Blockweave structure • Well suited to storing NFT metadata, historical records, and other information that must never disappear	• The initial upload cost per GB is higher than IPFS • Once written, data cannot be deleted, which is unsuitable for private information or correcting mistaken data • Because blockchain consensus is required, it is not suited to the instant delivery and updates that IPFS offers

3.3 Supplementary Note: Compatibility Between NFT Standards and Storage

Table 3-2 organizes the compatibility of combining CIP-25 / CIP-68 with IPFS / Arweave.

Combination	Compatibility / characteristics	Main weakness
CIP-25 x IPFS	Matches the assumption that metadata points to an external URI. The most common and highly compatible combination.	Weak persistence; a concern for high-value NFTs
CIP-25 x Arweave	CIP-25's non-updatable nature pairs well with Arweave's permanent storage. Advantageous for guaranteeing long-term value.	Write cost is higher than IPFS
CIP-68 x IPFS	Since the metadata body lives in the on-chain datum, it is less dependent on the persistence of an external URI. A natural fit for placing only large media such as images on IPFS.	Persistence of the image side still depends on pinning
CIP-68 x Arweave	Combines an updatable on-chain datum with permanent media storage, making it well suited to long-held NFTs such as RWAs and game assets.	Write cost becomes an issue if updates are frequent

Table 3-2 Compatibility comparison of NFT standards (CIP-25 / CIP-68) and storage (IPFS / Arweave)

3.4 Selected Storage and Rationale (Discussion)

Building on the comparison so far, this section considers which decentralized storage ChainCapsule should adopt. The axis for this decision is: "for a service like ChainCapsule, how critical would it be if recorded content were to disappear?"

ChainCapsule is a service premised on continuing to store records entrusted by users for what may be several decades. If data were ever lost from the storage destination, only the NFT — the "certificate" — would remain, while the content that matters most (the recorded text or photos) would be lost, a scenario that would undermine the very foundation of the service. From this standpoint, IPFS's characteristic that "data gradually disappears unless an operation to keep holding it (pinning) is performed" is a risk that ChainCapsule cannot afford to take lightly.

Arweave, on the other hand, is designed so that once a one-time fee is paid at upload, the data is retained, in theory, for over 200 years with no further cost. This is supported by an economic mechanism in which the majority (roughly 80%) of the fees users pay are accumulated into an "Endowment," which, on the assumption that storage unit costs will continue to fall over time, uses its investment returns to keep paying rewards to data custodians over the long term. The property that the network as a whole autonomously continues to protect data through this endowment mechanism, without needing any specific company or service to keep managing it, strongly matches ChainCapsule's underlying philosophy: "records should keep existing even if the administrator disappears."

It is also worth noting that Arweave hosts a decentralized computing platform called AO, which began operating in 2025, making it possible not only to store data but also to perform advanced computation on it. While this is an attractive technology as a mechanism for continuing a service even in the administrator's absence, at this stage ChainCapsule will use Arweave purely as storage and will not assume any use of AO.

Selected Storage	**Adopt Arweave (for the production environment)**
Rationale	**This project's core purpose is to "reliably preserve information recorded for the future until the specified date," so the permanence of storage is weighted even more heavily than for IPFS. Arweave's one-time payment model guarantees, in theory, permanent storage for over 200 years, and its economic incentive (the Endowment) provides continuity that does not depend on any specific operating entity. It is also highly compatible with CIP-25 (non-updatable), in that both offer a long-term guarantee that data will not change once finalized. For these reasons, Arweave is selected, with the highest priority placed on permanence.**

However, during the prototype stage, IPFS (via the Blockfrost IPFS service, offered by the same Blockfrost platform used for on-chain queries), which is easy to adopt, will be used on a provisional basis, prioritizing development cost and speed. The plan is to switch to Arweave at the time of migration to the production service; this phased migration approach is also reflected in the specification of Output C1, the "Prototype Basic Specifications Document."

NFT.Storage, which was originally considered as the IPFS provider, stopped accepting new uploads and pins on June 30, 2024, and has therefore been replaced by Blockfrost IPFS. This change concerns only the provider; the selection of IPFS for the prototype and Arweave for production is unchanged.

4. Summary

This report, Milestone 1's Output A1, investigates, compares, and selects between two technical elements that underpin ChainCapsule's trustworthiness: the NFT standard and decentralized storage.

As background, the report first clarifies the difference between Ethereum, where NFTs are issued primarily through smart contracts, and Cardano, where NFTs can be issued as native assets at low cost and low risk. It then explains Cardano's block and transaction structure (inputs, outputs, datum, metadata, and so on) and the metadata notation for CIP-25 and CIP-68, each illustrated with concrete examples. It also organizes how a Policy ID, which identifies the issuing entity of an NFT, is generated — whether derived from a wallet address or from a minting script.

For the NFT standard, CIP-25 and CIP-68 were examined together with a comparison table and their respective advantages and disadvantages. CIP-25 cannot have its metadata updated, but it can be implemented simply by writing it into the mint transaction, and it has the broadest wallet and marketplace support. CIP-68 offers the flexibility to update by rewriting the reference NFT's datum, but implementation is more complex because it requires managing two kinds of tokens: the User Token and the Reference NFT. Because ChainCapsule's core value lies in never changing a finalized record, priority was given to the fact that no update is needed, ease of implementation, and broad compatibility, and CIP-25 was adopted.

For decentralized storage, IPFS and Arweave were compared. IPFS makes tampering easy to detect through content addressing, but data can be lost if it is not pinned. Arweave, on the other hand, economically guarantees permanent storage for over 200 years through its Endowment mechanism, funded by a one-time payment. Given that reliably protecting records over a long period is fundamental to this service, Arweave is adopted for the production environment, with the highest priority placed on permanence. During the prototype stage, however, IPFS — which is easy to adopt — is used provisionally, prioritizing development cost and speed, with a planned phased migration to Arweave at

the time of production launch. The report also confirms that the combination of CIP-25 and Arweave is highly compatible, in that both offer a long-term guarantee for data that cannot be updated.

These selections are consistent with the project's existing concept, and are used as the technical premise for the next outputs, B1 and C1.